

The Broke Student's Breakaway—Optimizing Fixed-Gear Performance on a Limited Time and Budget

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Abstract—Fixed-gear cycling constrains cadence, which may change training responses. While polarized training—an ~80/20 split of low-intensity endurance and high-intensity intervals—is well supported in geared cycling, its efficacy for fixed-gear riders remains understudied. This study tested whether a structured nine-week program was associated with improved speed in one fixed-gear rider compared with a historical baseline of unstructured riding. Average speed (km/h) was the primary outcome. To reduce route-length bias, the main phase comparison restricted rides to a comparable 20–23 km distance band and used an Independent T-Test (W0 vs W1–W9). Weekly high-intensity minutes were aggregated from non-TT rides and modeled against weekly TT speed using OLS (HC3). A TT-only ANOVA compared flat versus clipless pedals on TT average speed. Within matched rides, the training phase was faster than baseline (mean 26.711 vs 21.772 km/h; $p = 1.29 \times 10^{-7}$; $\eta^2 = 0.481$). Weekly high-intensity minutes did not predict TT speed ($\beta = -0.034$; $p = 0.631$). Clipless pedals were associated with higher TT average speed ($p = 0.0098$; $\eta^2 = 0.698$). Ride-level correlations suggested HI share coincided with faster rides. Overall, structured training coincided with meaningful gains, whereas week-level dose–response effects were not detected in this small dataset.

Index Terms—Fixed-gear cycling, Polarized training, Single-subject design, Sports analytics, Aerobic efficiency, Interval training.

I. INTRODUCTION

Fixed-gear cycling, defined by a drivetrain lacking a freewheel mechanism, compels the rider to maintain continuous pedal motion, removing the ability to coast. Historically, professional road cyclists utilized fixed-gear bicycles during winter training blocks to refine pedal mechanics and cadence efficiency [1]. However, unlike modern road cycling, the fixed-gear mechanism strictly locks the rider's cadence to their speed. This introduces unique physiological constraints, as the rider cannot rely on gear reduction to manage terrain variability or utilize coasting for micro-recoveries. Consequently, effort distribution becomes a function of topography rather than rider choice, requiring a high degree of neuromuscular

control and sustained power output. In practice, this means fixed-gear riders must absorb speed changes through cadence alone, amplifying fatigue during surges and limiting deliberate pacing adjustments. In the Philippines, the popularity of fixed-gear cycling has surged, with sanctioned criterium races and local events increasingly including dedicated fixed-gear categories [2]. This discipline presents a steep learning curve compared to geared cycling. Without the mechanical advantage of variable gear ratios, athletes must rely entirely on physiological adaptation to manage varied race conditions. Furthermore, the typical absence of braking mechanisms in track-style frames increases technical handling requirements, distinguishing it fundamentally from standard road racing where speed modulation is mechanical rather than physical. As participation grows, riders often adopt improvised training approaches, creating a need for evidence-based guidance tailored to fixed-gear constraints.

Traditional cycling training regimes are well-documented and typically follow a "Polarized" model: a mix of low-intensity endurance (Zone 2) and high-intensity intervals (VO2 Max) [3]. However, these methodologies are predicated on the use of bicycles with variable gears, which allow riders to maintain optimal cadence and heart rate zones regardless of road gradient. It remains unclear if these standard regimes remain effective when applied to the restrictive mechanics of a fixed-gear bicycle. Therefore, this study addresses the following *primary research question*: **Can a structured, road-cycling-based training regimen be effectively transferred to fixed-gear cycling to yield statistically significant performance gains, despite the mechanical inability to optimize cadence?** By analyzing telemetry from a single-subject case study, this research seeks to validate low-cost, time-efficient training protocols for this mechanically distinct discipline.

II. LITERATURE REVIEW

To construct an effective training protocol for fixed-gear cycling, it is necessary to examine both physiological training theory and the mechanical constraints imposed by the drivetrain. This review synthesizes literature on polarized endurance training, the biomechanics of fixed-gear pedaling mechanics, and the expected time course of short-term physiological adaptation. Together, these works establish the theoretical basis and research gap addressed by this study.

The Physiology of Polarized Training

Modern endurance training theory is strongly influenced by the polarized training model, which organizes training intensity into a large proportion of low-intensity training (LIT) and a smaller proportion of high-intensity training (HIT). A systematic review by [4] analyzed endurance training literature retrieved from PubMed and SPORTDiscus and selected fourteen relevant experimental studies measuring $\dot{V}O_{2\max}$, $\dot{V}O_{2\text{peak}}$, and work economy. Across these studies, the most effective distribution consistently approximated 75–80% LIT and 15–20% HIT. The review reported significant improvements in aerobic capacity and work economy over short-term interventions, suggesting that endurance performance can be enhanced without relying heavily on threshold-intensity training. However, the authors also noted limited sample sizes and variability in study protocols, restricting broad generalization. Complementary findings are presented by [5], which compared polarized intensity training with other intensity distributions using empirical and comparative analysis. The study observed superior endurance performance in trained athletes using polarized methods, while still reporting measurable improvements among amateur participants. These results suggest that polarized training is not exclusive to elite athletes and may be effective even in recreational populations. Despite this, previous studies primarily examined athletes using conventional geared bicycles, leaving uncertainty regarding whether the same intensity distribution remains effective when cadence control is mechanically restricted, as in fixed-gear cycling.

Biomechanics of Fixed-Gear Cycling

Unlike geared bicycles, fixed-gear drivetrains mechanically couple crank rotation directly to wheel speed, preventing coasting and limiting cadence selection. The physiological consequences of cadence selection were examined by [6], who conducted graded laboratory testing on fourteen trained track cyclists at multiple pedaling rates. Using incremental exercise protocols and regression analysis of heart rate, oxygen uptake, lactate concentration, and power output, the study demonstrated that optimal cadence increases with exercise intensity, ranging from approximately 66 rpm at lower metabolic thresholds to over 84 rpm near maximal oxygen uptake. These findings indicate that efficient metabolic output depends on cadence matching

physiological demand. In fixed-gear cycling, however, riders cannot freely adjust cadence to metabolic state. Supporting this mechanical consideration, [7] conducted multistage graded cycling tests manipulating gear ratio and cadence while measuring gross efficiency and pedal force effectiveness through respiratory and force-reaction measurements. Results showed that increased gear ratio improved efficiency, whereas cadence changes had minimal influence on force effectiveness. Although performed on road bicycles, these findings suggest that drivetrain mechanics significantly influence energy expenditure. Consequently, fixed-gear riders may experience conflicts between physiologically optimal cadence and mechanically imposed cadence, creating uncertainty regarding whether traditional training prescriptions remain valid under constrained pedaling conditions.

Short-Term Adaptations in Amateur Cyclists

Because the present study evaluates a nine-week intervention, evidence of short-term adaptation is necessary. [8] reviewed low-volume high-intensity interval training protocols characterized by brief high-intensity efforts totaling less than fifteen minutes per session. Across clinical and healthy populations, HIIT produced improvements in cardiorespiratory fitness comparable to longer continuous training programs. Proposed mechanisms included enhanced mitochondrial function, insulin sensitivity, and cardiovascular efficiency. However, many of the reviewed studies emphasized health markers rather than sport-specific performance, leaving uncertainty regarding direct performance outcomes in cyclists. More directly related to cycling performance, [9] implemented a nine-week polarized training program among twenty-six mountain bike cyclists combining sprint interval training, high-intensity intervals, and endurance sessions. Laboratory sprint tests measured anaerobic power, lactate concentration, and reaction time before and after the intervention. Results showed increased anaerobic power and reduced performance errors following the polarized training period, although reaction time did not significantly change. These findings demonstrate that measurable physiological adaptation can occur within a two-month training cycle. Nevertheless, prior work was conducted in geared cycling disciplines, and the effect of similar training under the continuous pedaling demands of fixed-gear cycling remains unexplored.

Collectively, the reviewed literature establishes that polarized training improves endurance performance, cadence directly affects metabolic efficiency, and measurable adaptations can occur within short training cycles. However, no study evaluates these principles under the mechanical constraints of fixed-gear cycling, necessitating the experimental methodology this study proposes.

III. METHODOLOGY

This chapter describes the procedures used to evaluate whether shifting from unstructured fixed-gear riding to a structured, time-efficient training regimen produces measurable performance gains. The study relies on personal cycling telemetry (e.g., speed, distance, and time) recorded through Strava and uses a single-subject, quasi-experimental pre/post design with a historical control group. The control dataset consists of approximately 200 unstructured fixed-gear rides collected over multiple years and exported from the Strava data archive (FIT files), while the experimental dataset is collected during a structured training intervention using the same recording workflow with additional heart-rate logging. To reduce route variability and ensure replicable comparison across weeks, performance is evaluated using a standardized weekly 20 km flat highway loop as the primary test route.

While cycling is naturally an aerodynamic sport, wind resistance will not be factored in this study as to mimic actual race conditions that can happen both in sanctioned races and in unsanctioned ones. Therefore cycling aerodynamics with the bike frame, the rider's clothes, and positioning are not tracked. To keep the performance tracking consistent, the rider cannot slipstream on other vehicles and are only allowed to ride solo. This is to mimic "break aways" where riders move away from the peloton to advance, some with other riders while some do it alone.

A. Participants

The participant is the researcher (the student) acting as the sole subject of the study. The subject is a 22-year-old male college student with three years of fixed-gear cycling experience, previously riding primarily during free time rather than training specifically for speed. During the intervention, body mass ranged from 86–89 kg and height was 1.83 m. As this is a single-subject study, the results are intended to describe within-subject performance changes rather than generalize to all cyclists.

B. Apparatus

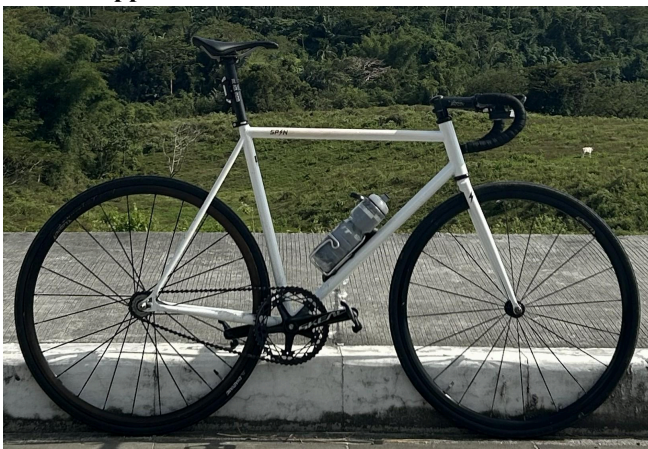


Image 1. Subject's Bike

The bicycle used is a Spin 2025 chromoly steel frameset (56 cm). Build components include: Kocevlo titanium-railling saddle, Roadnado 400 mm seatpost, Wake 110 mm -17° stem, LaBici 40 mm drop bars, Skeace square-tapered crankset, Legend rims, Novatec hubs, Pillar spokes, and 28c tires. Total bike mass is approximately 9 kg. From Weeks 1–6, the bike used Pizz flat pedals with Sackit straps; from Weeks 7–9, Shimano RS500 clipless pedals with Tabolu Road cleat shoes were used.

C. Experimental Design and Procedures

The training structure is adapted from a road cycling regimen popularized by content creator Angelo Bikerdude [10]. While originating from community practice, the protocol aligns empirically with Polarized Training principles. The regimen distributes intensity across three distinct modalities (detailed in Table 1) designed to balance aerobic conditioning with high-intensity structural adaptations, simulating the physical demands of competitive racing within a time-constrained schedule.

Routine	Purpose	Qualifications
Zone 2 Climbing	To enhance mitochondrial density and fat oxidation without accumulating significant fatigue.	>75% of moving time must be spent in Heart Rate Zone 2 (60–70% of Max HR).
Interval Training	To increase VO2 Max ceiling and improve lactate clearance during supra-threshold efforts.	3–5 Sets of maximum effort sprints (120s work) followed by active recovery (120s rest).
Time Trial Test	To benchmark performance and simulate the sustained threshold power required for "breakaways."	20 km sustained maximum effort on the standardized route. No drafting or stops.

Table 1. Weekly Training Routine and Qualifications.

D. Study Site and Standardization

To minimize environmental confounding, performance benchmarking (Time Trials) was conducted on a standardized 20 km flat highway loop along the Maragondon Bypass Road (Cavite, Philippines). Selected for its negligible elevation change (<50 m gain) and consistent pavement quality, this route isolated physiological output as the primary determinant of speed variance. Time trials were synchronized by time-of-day to control for wind resistance and traffic density. Conversely, Zone 2 endurance efforts were situated on the Naic–Indang Road. This route was selected for its consistent, moderate gradient (<250 m total gain; avg 3%, max 7%), offering a physiological challenge compatible with a single-speed drivetrain where mechanical gear reduction is impossible.

E. Data Collection Methods

Data were collected for each ride performed during the training weeks, with historical rides extracted from Strava to serve as baseline. The subject completed an average of four rides per week during the experimental period. Each ride was labeled using standardized names to indicate training type, including: “DS Zone 2,” “DS INTERVAL,” “DS TT TEST,” and “DS Long ride.” This naming convention ensured consistent grouping and weekly analysis of ride categories.

Specific data collected:

- id: Unique ride identifier from Strava.
- name: Ride title/name as recorded in Strava.
- distance: Total ride distance (kilometers).
- moving_time: Total time in motion (minutes).
- elapsed_time: Total ride duration including stops (minutes).
- total_elevation_gain: Total elevation gained during the ride (meters).
- start_date: Ride timestamp (date and time the activity started).
- average_speed: Mean speed over the ride (km/h).
- max_speed: Highest recorded speed during the ride (km/h).
- average_temp: Average ambient temperature recorded during the ride (°C).
- elev_high: Highest elevation reached during the ride (meters).
- elev_low: Lowest elevation reached during the ride (meters).
- pr_count: Number of personal records detected by Strava for the ride (automatically computed by the platform).
- avg_hr: Average heart rate during the ride (bpm).
- max_hr: Maximum heart rate recorded during the ride (bpm).
- min_hr: Minimum heart rate recorded during the ride (bpm).
- cleats: Pedal system indicator (0 = flat pedals, 1 = clipless/cleats).

Tools / apps used

1. Ride recording: a GPS cycling computer (XOSS G2+) recorded ride telemetry (time, distance, speed, elevation), which was then synced to Strava.
2. Historical baseline retrieval: historical rides were obtained from Strava using a custom export workflow (via API/script) and stored for analysis.
3. Heart-rate recording (experimental rides): heart rate was measured using Apple Watch SE and manually entered when needed because the watch export format did not directly match the cycling computer export format.

F. Operational Definitions and Feature Engineering

To ensure replicability, the following operational definitions were used:

- Historical baseline (control): all past unstructured fixed-gear rides exported from Strava prior to the intervention.
- Experimental rides: rides performed during the structured training period and labeled using the “DS ____” naming convention.
- Primary performance outcome: average_speed (km/h) on the weekly 20 km flat highway loop, used as the standardized performance test.
- Training frequency: average number of rides completed per week during the intervention (target ≈ 4).
- Pedal condition (cleats): a binary indicator where 0 = flat pedals and 1 = clipless pedals (Weeks 7–9).

G. Data Cleaning and Preprocessing

Before analysis, the dataset was cleaned to ensure consistency across historical and experimental rides. First, units were standardized so that all time variables were represented in minutes and all speed and distance variables used the same measurement system (km/h and kilometers). Date and time values were converted into a consistent datetime format to allow weekly grouping and time-based comparisons. Missing values were expected for heart-rate variables in historical rides since heart-rate logging was only available during the experimental phase; therefore, analyses comparing historical vs experimental performance were conducted using variables available in both datasets, while heart-rate analyses were restricted to experimental rides only. Outliers and improbable values (e.g., unusually high speeds caused by GPS errors or rides with excessive stops) were screened during exploratory analysis and either corrected where possible or excluded when they could not be validated.

Engineered Features

- **z1_minutes ... z5_minutes:** Minutes spent in heart-rate Zones 1–5 (converted from “HH:MM:SS” time strings).
- **hi_minutes:** Total high-intensity minutes, computed as z4_minutes + z5_minutes.
- **total_zone_minutes:** Total minutes across all zones, computed as z1 + z2 + z3 + z4 + z5 (minutes).
- **hi_share:** Proportion of high-intensity work, computed as hi_minutes / total_zone_minutes.
- **week:** Week label (W0 for historical baseline; W1–W9 for training weeks) extracted from ride naming/metadata for grouping.
- **is_training_phase:** Binary indicator used for modeling (0 = W0 historical, 1 = W1–W9 training).

- **efficiency_factor:** Efficiency proxy computed as $\text{average_speed} / \text{avg_hr}$ (computed only when heart rate is valid).
- **days_since_last_ride:** Recovery proxy computed from the difference in days between consecutive start_date values (chronological order).
- **fast_ride:** Binary indicator for “fast rides,” defined as $\text{average_speed} > \text{median}(\text{average_speed})$ for chi-square analysis.
- **tt_flag / tt_speed:** Time-trial marker based on ride labels (“DS TT TEST”), with tt_speed representing the mean speed of TT rides per week for weekly modeling.

H. Statistical Analysis

All statistical analysis was performed in Python using clean ride-level telemetry. The primary outcome was average_speed (km/h). To control for route-length bias, the primary baseline comparison (Historical vs. Training) was restricted to a matched distance band of 20–23 km.

Exploratory Analysis: Distribution and trend patterns were assessed using boxplots (speed variability), time-series line plots (efficiency trends), and correlation heatmaps (variable associations). These visualizations verified outlier presence and intensity adherence prior to inferential testing.

Training effect on speed (H1): To test for performance improvements, a one-way ANOVA compared average_speed between Historical (W0) and Training (W1–W9) phases within the matched 20–23 km band. Statistical significance was reported using the F-statistic, p-value, and effect size (η^2), with group means inspected to confirm directionality..

Training structure and intensity relationships: Analysis of heart-rate derived metrics (e.g., efficiency_factor, hi_share) was restricted to the training phase (W1–W9). Relationships between physiological load and speed were quantified using descriptive statistics and Pearson correlation coefficients.

Does intensity predict time-trial performance (H2): To link training volume to results, an OLS regression with HC3 robust standard errors modeled weekly tt_speed against weekly_hi_minutes (aggregated from non-TT rides to avoid circularity). A Spearman rank correlation was used as a robustness check to account for the small weekly sample size and potential non-linearity.

Equipment Effect (H3): The impact of clipless pedals was evaluated using a one-way ANOVA on average_speed, restricted strictly to standardized TT rides. Results were interpreted using effect size (η^2) and group means to contextualize findings within the specific constraints of the dataset.

I. Hypotheses and Inference Questions

Hypothesis 1 (Training effect on speed)

- **H0:** There is no significant difference in mean average_speed between historical rides (W0) and training rides (W1–W9).
- **Ha:** Mean average_speed during training rides (W1–W9) is higher than the historical baseline (W0).

Hypothesis 2 (weekly_hi_minutes as predictor of speed).

- **H0:** The regression coefficient of weekly_hi_minutes is equal to zero ($\beta = 0$).
- **Ha:** The regression coefficient of weekly_hi_minutes is greater than zero ($\beta > 0$).

Hypothesis 3 (Clipless vs flat pedals).

- **H0:** Switching to a clipless system does not improve performance (no increase in speed and no reduction in moving_time for comparable rides).
- **Ha:** Switching to a clipless system improves performance (higher speeds and/or lower moving_time for comparable rides).

IV. RESULTS

This chapter presents the results of the data analysis comparing historical fixed-gear rides (W0) to rides recorded during a structured nine-week training phase (W1–W9). Findings are reported using descriptive summaries, visualizations, correlation analysis, and hypothesis-testing outputs focused on performance during standardized test rides and training structure metrics.

Overview of Combined Dataset

The raw combined dataset contained 23 columns and 265 rows, with 225 rows labeled Week 0 (pre-training) and 40 rows labeled Weeks 1–9 (training phase). After cleaning and feature engineering, the dataset contained 263 rows and 35 columns. The resulting dataset was used to generate the figures and statistical tests presented below.

Figures



Figure 1. Heatmap correlation of engineered features.

Figure 1 correlation heatmap summarizes pairwise relationships among engineered variables and performance metrics. Average speed shows a positive association with high-intensity exposure, with stronger correlation to high-intensity share ($r = 0.67$) than high-intensity minutes ($r = 0.46$). Average heart rate is also positively related to

high-intensity variables ($r \approx 0.53\text{--}0.62$). Days since the last ride shows weak correlations with most variables ($r \leq 0.28$). Efficiency factor shows a strong positive correlation with average speed ($r = 0.87$), consistent with its computed definition.

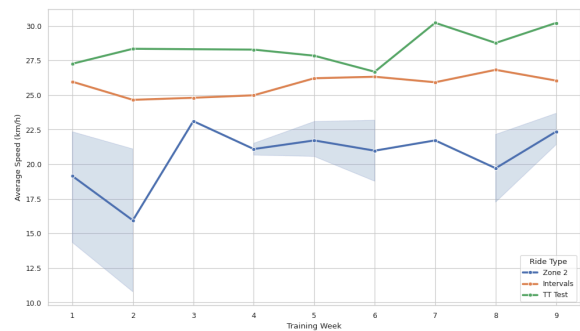


Figure 2. Speed trend by ride type.

Figure 2 line chart tracks weekly average speed by ride type (Zone 2, Intervals, TT Test) across Weeks 1–9. TT Test rides remain the fastest each week, interval sessions remain mid-range, and Zone 2 rides remain the slowest. TT Test speeds remain in the high-20 km/h range with minor week-to-week variation. Interval speeds remain relatively stable around the mid-20 km/h range. Zone 2 speeds show greater variability early in the training block and appear more stable in later weeks.

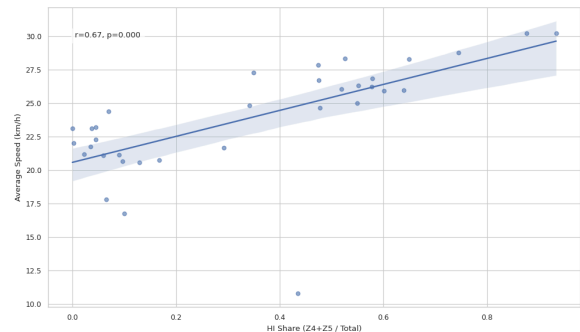


Figure 3. High-intensity share vs speed.

Figure 3 scatterplot shows the relationship between high-intensity share (proportion of time in Z4–Z5) and average ride speed. The fitted trend line indicates an upward relationship, where higher high-intensity share values are associated with higher average speeds. The correlation is moderately strong ($r = 0.67$) and statistically significant ($p < 0.001$).

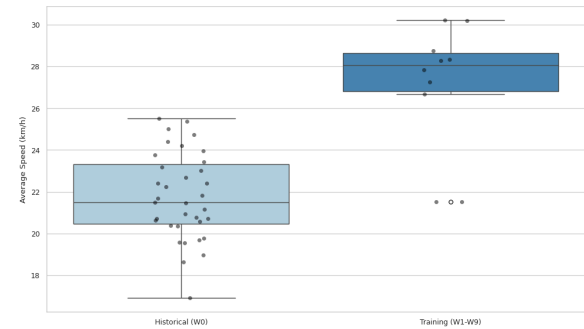


Figure 4.Speed of 20 km rides in Week 0 vs Weeks 1–9.

Figure 4 boxplot compares the distribution of average speed on the standardized 20 km route between historical baseline rides (W0) and training rides (W1–W9). The training distribution is shifted upward relative to baseline, with higher central tendency and a tighter cluster of values. Baseline rides show wider dispersion and include more low-speed observations.

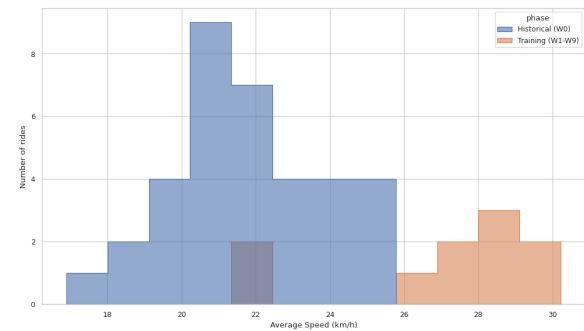


Figure 5. Average speed histogram (matched 20–23 km rides).

Figure 5 histogram compares average speed distributions for matched 20–23 km rides between historical baseline (W0) and the training phase (W1–W9). Baseline speeds cluster largely in the low-20 km/h range, while training speeds cluster in the high-20 km/h range, showing limited overlap between phases.

Tables

Variable	Mean	Median	SD
Average speed (km/h)	23.301	23.095	4.130
HI share (Z4+Z5 / total)	0.340	0.346	0.279
HI minutes (Z4+Z5)	12.698	12.442	11.111
Average HR (bpm)	155.457	153.000	13.312
Efficiency factor (km/h per bpm)	0.154	0.156	0.024

Table 2. Training Block Descriptive Statistics (W1–W9).

Table 2 summarizes mean, median, and standard deviation for key training-phase metrics: average speed (mean = 23.301 km/h; SD = 4.130), high-intensity share (mean = 0.340; SD = 0.279), high-intensity minutes (mean = 12.698; SD = 11.111), average heart rate (mean = 155.457 bpm; SD = 13.312), and efficiency factor (mean = 0.154; SD = 0.024).

Validating H1 using ANOVA

phase	n	mean	sd	median
Historical (W0)	35	21.772	2.076	21.49
Training (W1–W9)	9	26.711	3.109	27.84

Table 3. Descriptive Statistics of Average Speed by Phase (matched 20–23 km rides)..

For matched-distance rides, the historical baseline (W0; $n = 35$) recorded mean average speed of 21.772 km/h (SD = 2.076; median = 21.49), while the training phase (W1–W9; $n = 9$) recorded mean average speed of 26.711 km/h (SD = 3.109; median = 27.84). The mean difference between phases is approximately 4.94 km/h in favor of training rides within the 20–23 km band.

	sum_sq	df	F	PR(>F)
C(phase)	217.595337	1.0	39.840037	0.000001
Residual	234.854187	43.0	NaN	NaN
ANOVA result = $F(1, 43) = 39.8400$, $p = 1.2858e-07$, $\eta^2 = 0.481$				

Table 4. Weekly TT outcomes and weekly intensity inputs (W1–W9, excluding Week 3 with no TT ride).

Weekly TT test outcomes (tt_speed) and weekly training inputs (weekly_hi_minutes and weekly_z2_minutes) were paired for Weeks 1–2 and 4–9 ($n = 8$ usable weeks). TT speed remained within a relatively narrow range across weeks (mean = 28.449 km/h; SD = 1.270), while weekly high-intensity minutes varied more widely (mean = 35.323 minutes; SD = 11.641).

A one-way ANOVA tested whether average speed differed by phase (historical W0 vs training W1–W9) for matched 20–23 km rides. The effect of phase was significant, $F(1, 43) = 39.8400$, $p = 1.2858 \times 10^{-7}$, with a large effect size $\eta^2 = 0.481$. **Based on the ANOVA outcome and the higher training mean in Table 3, H0 was rejected for H1.**

Validating H2 using OLS Regression

Variable	Mean	SD	Min	Max
TT speed (km/h)	28.449	1.270	26.680	30.220
Weekly HI minutes (Z4+Z5)	35.323	11.641	18.883	50.517
Rest days before TT	1.736	1.101	0.744	3.858

Table 5. Summary statistics of weekly time-trial speed and weekly training inputs.

Table 5 summarizes the weekly dataset used to evaluate H2 by pairing each week's TT test outcome with training inputs accumulated from non-TT rides. Across the eight usable weeks, TT speed was relatively stable (mean = 28.449 km/h, SD = 1.270; range = 26.680–30.220), while weekly high-intensity exposure showed greater variability (mean = 35.323 minutes, SD = 11.641; range = 18.883–50.517). Rest days before the TT averaged 1.736 days (SD = 1.101; range = 0.744–3.858), indicating that recovery timing also differed across weeks. Using this weekly structure, an OLS regression with HC3 robust standard errors ($\text{tt_speed} \sim \text{weekly_hi_minutes}$; $n = 8$) found that weekly HI minutes did not significantly predict TT speed ($\beta = -0.0341$, $p = 0.631$, $R^2 = 0.098$). Therefore, **fail to reject H0 for H2**, as there is no statistical evidence in this dataset that higher weekly high-intensity minutes are associated with higher TT speed.

Validating H3 using One-way ANOVA

	sum_sq	df	F	PR(>F)
C(cleats)	7.879687	1.0	13.86862	0.009805
Residual	3.409000	6.0	NaN	NaN
ANOVA result: $F(1, 6) = 13.8686$, $p = 9.8052e-03$, $\eta^2 = 0.698$				

Table 6. One-way ANOVA results for TT-only average speed by pedal system (flat vs clipless).

A one-way ANOVA was conducted to test whether TT test average speed differs between flat pedals and clipless pedals. The analysis shows a statistically significant effect of pedal system on TT average speed, $F(1, 6) = 13.8686$, $p = 0.009805$, with a large effect size ($\eta^2 = 0.698$), indicating that a substantial proportion of variance in TT average speed is associated with pedal system. Since the result is significant ($p < 0.05$) and the clipless group showed higher mean TT average speed in the descriptives, **reject H0 and support Ha for H3** (clipless pedals are associated with higher TT average speed).

V. DISCUSSION

This study examined whether shifting from unstructured fixed-gear riding to a structured nine-week program was associated with measurable performance changes. Using matched-distance rides (20–23 km) to improve comparability between phases, the training period showed higher average speeds than the historical baseline (H1). This finding was visible in both distribution plots (boxplot and histogram) and supported by a one-way ANOVA with a large effect size, indicating that phase membership (historical vs. training) accounted for a substantial portion of the observed speed variation within the matched-distance subset. Together, these outcomes suggest that the structured training block coincided with a meaningful improvement in sustained pace for comparable rides.

A. Interpretation of Results

This project evaluated whether a shift from unstructured fixed-gear riding to a structured nine-week program was associated with measurable performance changes. The primary finding (H1) was that average speed increased during the training phase compared with the historical baseline when rides were restricted to a matched-distance band (20–23 km). The upward shift seen in the distribution plots and the significant one-way ANOVA with a large effect size indicates that the training phase is strongly associated with higher sustained speed on comparable-length rides. In practical terms, this suggests the program coincided with improved pacing capacity and/or improved ability to hold speed over a full effort.

Within the training block, ride-type trends also supported that the program produced distinct intensity outputs. TT tests were consistently the fastest sessions, interval rides remained mid-range, and Zone 2 rides were the slowest, which aligns with how each session type was intended to be performed. In addition, the relationship between intensity exposure and performance appeared clearly at the ride level: higher high-intensity share tended to occur alongside higher average speeds. This pattern is consistent with the idea that rides with more Z4–Z5 exposure reflect harder efforts and therefore produce higher speeds during those sessions.

However, the weekly-level intensity hypothesis (H2) was not supported. When high-intensity minutes were aggregated by week and used to predict that week's TT speed, the regression was non-significant and explained little variance. This result suggests that weekly TT performance did not move in a predictable way with weekly high-intensity minutes in this dataset. A likely explanation is that TT speed varied within a relatively narrow range across weeks while training inputs varied more widely, and with only a small number of usable TT weeks, the analysis had limited sensitivity to detect a week-to-week dose-response effect. Real-world variation (wind, traffic, fatigue, pacing, and day-to-day readiness) can also introduce noise that weakens weekly predictive relationships.

For equipment effects (H3), restricting analysis to TT-only rides improved comparability by focusing on standardized efforts. The ANOVA results indicate that the pedal system was associated with differences in TT average speed, with clipless rides showing higher sustained performance than flat-pedal rides and a large effect size. This suggests that clipless pedals may provide an advantage that shows up primarily in sustained output (holding speed across the full effort), rather than in short peak-speed moments.

B. Comparison of Related Work

The findings are broadly consistent with common endurance-training principles reported in cycling and exercise science literature: structured programs that separate session goals (low-intensity base work, intervals, and periodic testing) are typically used to improve sustainable speed and time-trial performance over time. The ride-level association between high-intensity exposure and speed is also consistent with the expectation that harder sessions produce higher speeds during those efforts.

At the same time, the absence of a significant weekly intensity → TT speed relationship (H2) is not unusual in short training blocks with small sample sizes. Prior work and coaching literature often emphasize that performance responses can lag behind training dose, vary by recovery, and be sensitive to conditions—factors that can mask week-to-week statistical relationships even if training is beneficial overall. In other words, the results indicate “training phase faster than baseline,” but does not clearly support “more HI minutes this week = faster TT this week.”

C. Limitations

Several limitations affect how confidently the results can be generalized:

1. Single-subject design ($n = 1$): Results reflect one rider's response and may not generalize to others.
2. Historical control (non-randomized): Differences between W0 and W1–W9 could partially reflect confounders such as season, schedule, traffic patterns, route familiarity, or gradual fitness changes unrelated to the program.
3. Small inferential samples for H2 and H3: Weekly TT modeling used a few weeks, and TT-only pedal comparisons used few clipless TT rides, making results sensitive to outliers and conditions.
4. Measurement and environment noise: GPS drift, stop-and-go traffic, wind, and road conditions can affect speed independent of fitness.
5. Missing physiological comparability across phases: Heart-rate data was not available for historical rides, limiting cross-phase physiological interpretation and making “efficiency” comparisons mainly training-only.

D. Recommendations and Future work

To strengthen the experimental design and model interpretability in future studies, the following refinements are recommended:

- **Standardization & Baseline:** Replace historical aggregation with a dedicated pre-intervention baseline block. Strict controls on Time Trial (TT) start times and environmental logging (wind/traffic) are essential to minimize external variance.
- **Study Duration:** Extending the observation window to 12–16 weeks would increase statistical power and mitigate the impact of week-to-week noise on regression models.
- **Feature Expansion:** Incorporate recovery variables (e.g., Sleep, RPE) and granular intensity metrics (e.g., interval bout counts rather than aggregate high-intensity minutes) to better capture physiological readiness and training quality.
- **Balanced A/B Testing:** Future equipment comparisons (e.g., cleats vs. flats) should utilize a balanced sample size design to ensure statistical validity and prevent imbalance-driven bias.

VI. CONCLUSION

This study aimed to determine whether the subject's shift from unstructured fixed-gear riding to a structured nine-week training program (W1–W9) was associated with measurable improvements in cycling performance compared with a historical baseline (W0). Using Strava-derived telemetry and a matched-distance comparison (20–23 km) to reduce route-length bias, the analysis found clear evidence that average speed during the training phase was higher than the historical baseline. The one-way ANOVA for H1 indicated a statistically significant phase effect with a large effect size, supporting the conclusion that the structured block coincided with faster sustained riding on comparable-distance efforts.

Across the training period, performance patterns were consistent with the intended training structure: TT rides produced the highest weekly speeds, interval sessions remained in the mid-range, and Zone 2 rides produced the lowest speeds. At the ride level, higher high-intensity exposure—especially high-intensity share—was associated with higher average speed. However, when intensity was evaluated at the weekly level (H2), weekly high-intensity minutes did not significantly predict weekly TT speed, indicating that week-to-week variation in TT outcomes was not explained by weekly HI volume in this dataset. For equipment effects (H3), TT-only comparisons showed that clipless pedals were associated with higher TT average speed, suggesting a measurable benefit for sustained performance during standardized test efforts.

From analyzing the dataset, the subject learned that adopting a structured routine—particularly separating session types and maintaining a consistent TT testing route—aligned with improved sustained pace compared with the historical baseline. The findings also suggest that phase-level comparisons and standardized tests captured performance changes more clearly than week-to-week modeling of intensity minutes and TT outcomes, especially given the small number of usable weeks and real-world variability (e.g., wind, traffic, pacing). In practical terms, the results support continuing the use of a consistent test route, preserving clear session categories (Zone 2, intervals, TT), and prioritizing sustained-speed outcomes as the main indicator of improvement. Overall, the study provides evidence that the subject's structured training phase was associated with higher average speed on comparable rides, while also showing that weekly high-intensity minutes alone did not explain week-level TT performance in this dataset.

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